

Discrete Delta Hedging under Black–Scholes

Convergence, transaction costs, volatility misspecification and model risk

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Abstract

We study, by Monte–Carlo simulation, the performance of the discrete-time delta-hedging strategy for a short option position in the Black–Scholes market. Using an *exact* geometric-Brownian-motion scheme, a fully vectorised hedging engine and common random numbers, we quantify how the hedging error depends on the rebalancing frequency, the drift, transaction costs, the hedging volatility, and the choice of underlying model. Every statistic is reported with its Monte–Carlo standard error. We find that (i) the mean hedging error is exactly zero under the pricing measure, so the object of interest is its *dispersion*, which collapses as $n^{-1/2}$ (fitted exponent -0.468 ± 0.009); (ii) the dispersion *does* depend on the drift, contrary to a common simplification; (iii) digital options do not obey the $1/\sqrt{n}$ law (exponent ≈ -0.24); (iv) transaction costs induce a cost/variance trade-off with an interior optimal frequency, improved by Leland’s adjusted volatility; (v) the hedging error is governed by realised-vs-implied variance through the gamma-P&L identity, validated against the engine to three decimals; and (vi) under stochastic volatility and jumps the Black–Scholes delta hedge leaves fat-tailed residual risk. All results are reproducible from the accompanying `deltahedge` package and its test suite.

1 Framework

The risky asset follows a geometric Brownian motion and the money-market account grows at the risk-free rate r :

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad dB_t = r B_t dt.$$

We sell a European option priced $V(t, S_t)$ and hold the self-financing hedge $\pi_t = \Delta_t S_t + \beta_t B_t$ with $\Delta_t = \partial_S V$. By Itô’s formula and the Black–Scholes PDE, $d\pi_t = \Delta_t dS_t + \beta_t dB_t$, so in *continuous* time the hedge replicates the payoff exactly, $\pi_T = V(T, S_T)$. In *discrete* time we rebalance only at n dates $t_k = kT/n$, leaving the hedging error

$$\varepsilon_n = \pi_n(T) - \text{payoff}.$$

The drift μ never enters the hedge — the engine observes only realised prices — and under the pricing measure ($\mu = r$) the discounted self-financing portfolio is a martingale, so

$$\mathbb{E}[\varepsilon_n] = 0 \quad \text{exactly, at every } n.$$

The mean is therefore *not* the quantity to study; the dispersion of ε_n is. This single observation reframes the entire analysis.

Baseline. $r = 10\%$, $\sigma = 20\%$, $S_0 = 100$, $K = 90$, $T = 0.25$ y; $N = 50,000$ Monte–Carlo paths; frequencies $n \in \{0, 1, 3, 12, 84\}$ (and 252 for the digital). We use the calendar-day convention (1 year = 12 months = 48 weeks = 336 days), so $n = 84$ is *calendar*-daily over a quarter; a real trading quarter has ≈ 63 trading days, a point we keep explicit rather than hide.

2 Methodology

Three methodological choices distinguish this study from a textbook exercise.

1. **Exact GBM, not Euler.** Paths use the exact log-Euler scheme $S_{k+1} = S_k \exp[(\mu - \frac{1}{2}\sigma^2)\delta t + \sigma\sqrt{\delta t}Z_k]$, which is exact for GBM. The arithmetic Euler–Maruyama scheme injects an $O(\delta t)$ bias that would masquerade as a “hedging error” at coarse n ; the exact scheme removes it (and any risk of negative prices).
2. **Common random numbers.** All frequencies (and all drifts) are driven by a single fine Brownian motion, aggregated exactly to each coarser grid, so cross-frequency comparisons are not blurred by Monte–Carlo noise.
3. **Signal vs. noise.** Every estimate is reported with its standard error $\text{SE} = \text{std}/\sqrt{N}$ and a 95% confidence interval. A number is only called “an effect” once it is statistically resolved.

3 Convergence of the call hedge ($\mu = r$)

Table 1 reports the hedging error by frequency. The mean is statistically indistinguishable from zero everywhere (every p -value is large, every 95% CI contains 0), exactly as the martingale property predicts. The real, resolved effect is the collapse of the standard deviation, from 9.59 (no hedge) to 0.21 (calendar-daily): a $\sim 98\%$ reduction.

Table 1: Call hedging error, $N = 50,000$, common random numbers across n . The mean is consistent with zero at every frequency; the dispersion is the signal.

n	frequency	mean	SE	95% CI	std	$p(\text{mean} = 0)$
0	No hedging	−0.0710	0.0429	[−0.155, +0.013]	9.5853	0.10
1	Single (static)	−0.0027	0.0066	[−0.016, +0.010]	1.4742	0.68
3	Monthly	−0.0028	0.0044	[−0.011, +0.006]	0.9762	0.52
12	Weekly	−0.0016	0.0023	[−0.006, +0.003]	0.5247	0.49
84	Calendar-daily	+0.0004	0.0009	[−0.001, +0.002]	0.2058	0.66

The $1/\sqrt{n}$ law, fitted properly. A single ratio $\sigma_{n=1}/\sigma_{n=84} \approx 7.2$ is *not* evidence of $1/\sqrt{n}$ (the law predicts $\sqrt{84} \approx 9.2$). Fitting $\log \sigma = a + b \log n$ on $n \geq 3$ — excluding the static $n = 1$, which performs no intermediate rebalancing and lies outside the asymptotic regime — gives

$$b = -0.468 \pm 0.009 \quad (95\% \text{ CI } [-0.486, -0.450]),$$

cleanly consistent with the theoretical $-\frac{1}{2}$. The diagnostic $\sigma\sqrt{n}$ rises from 1.47 at $n = 1$ to a ≈ 1.89 plateau for $n \geq 3$, which is precisely why $n = 1$ must be excluded.

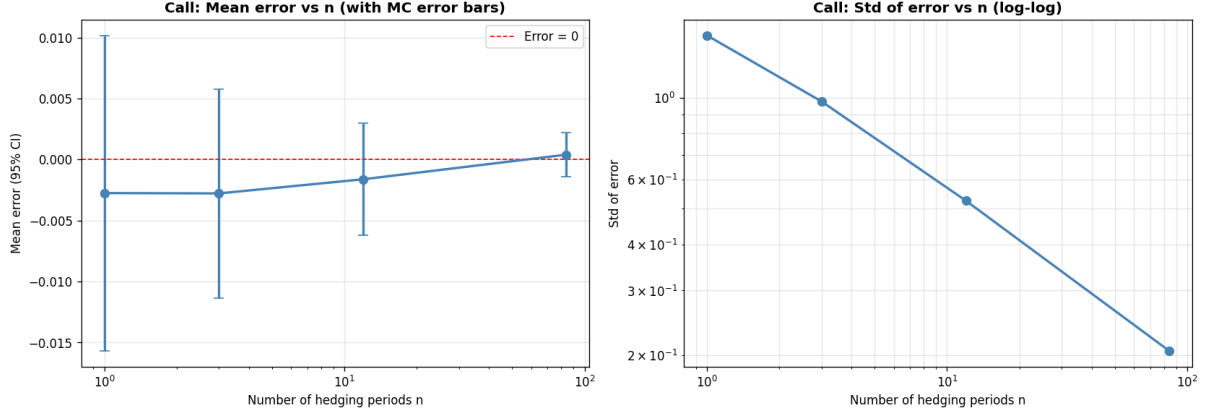


Figure 1: Left: mean error with 95% Monte-Carlo error bars (all crossing zero). Right: standard deviation on a log-log axis — a clean power law with slope ≈ -0.47 .

4 Does the drift μ matter?

A common simplification states that the hedging-error variance is controlled by the rebalancing frequency and not by the drift. This is *false*. Using common random numbers across μ , the $n = 84$ dispersion falls monotonically as μ rises (Table 2): a higher drift carries paths deep in-the-money into the low-gamma region, where there is less gamma-P&L to mis-hedge. The mean stays zero throughout — drift moves the *variance*, not the *bias*.

Table 2: Standard deviation of the call hedging error at $n = 84$ versus drift μ (CRN across μ).

μ	-0.10	0.00	0.05	0.10	0.15	0.30
std	0.2545	0.2325	0.2192	0.2058	0.1941	0.1578

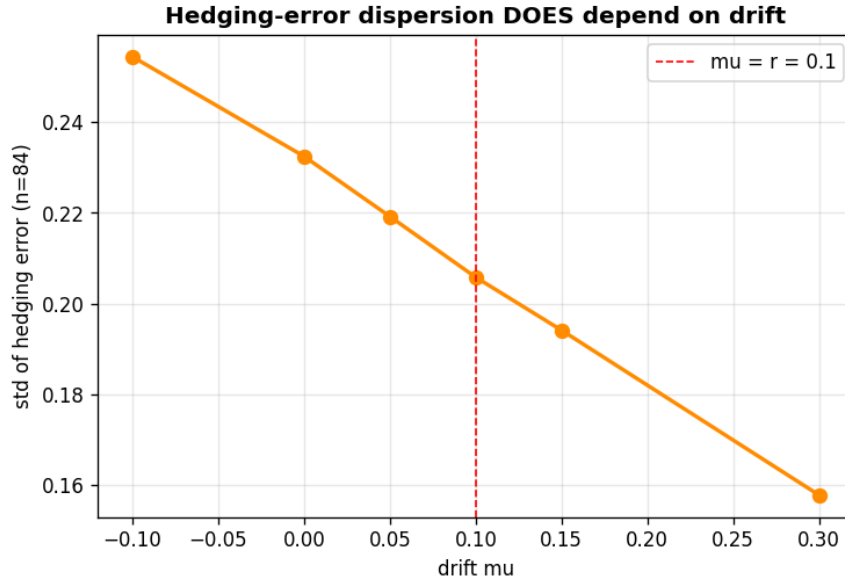


Figure 2: Hedging-error dispersion decreases with the drift — it is not drift-free.

5 Digital option

The cash-or-nothing call (payoff = $\mathbf{1}\{S_T \geq K\}$) is genuinely harder to hedge. Two findings are real (Table 3): the error distribution is *bimodal* (clusters at payoff 0 vs. 1), and the std-scaling exponent is only ≈ -0.24 — the digital error does *not* obey $1/\sqrt{n}$. Even $n = 252$ leaves a std of ~ 0.067 , because the delta spikes near the strike close to expiry. (By contrast, the original project’s “non-monotonic mean error” was pure noise: every $p > 0.05$.)

Table 3: Digital hedging error, $N = 50,000$. Means are all consistent with zero; the dispersion shrinks far more slowly than for the vanilla call.

n	frequency	mean	SE	std
0	No hedging	+0.0001	0.0014	0.3066
1	Single	−0.0005	0.0011	0.2555
3	Monthly	+0.0000	0.0009	0.1990
12	Weekly	+0.0002	0.0006	0.1436
84	Calendar-daily	−0.0004	0.0004	0.0893
252	$\sim 4\times$ /trading day	−0.0004	0.0003	0.0673

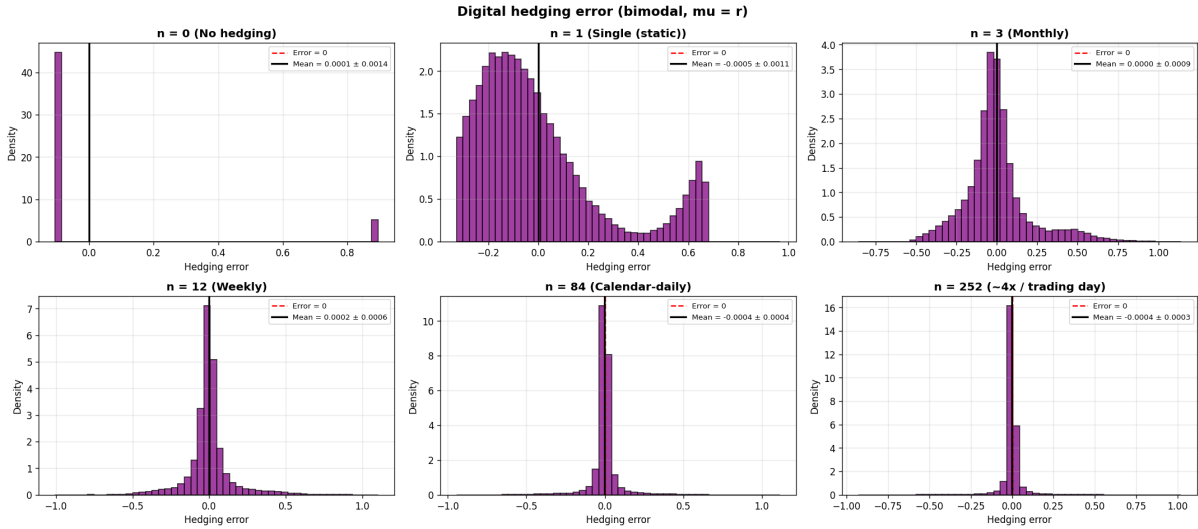


Figure 3: The digital hedging error is bimodal at every frequency, reflecting the discontinuous payoff.

6 Transaction costs and the Leland adjustment

With a proportional cost κ per unit of underlying traded, more frequent rebalancing cuts the hedging-error dispersion but raises the expected cost drag. The two effects trade off, so a tail-risk measure such as the 5% CVaR of the loss has an *interior optimum*: at $\kappa = 0.1\%$ the CVaR-optimal frequency is $n^* \approx 168$ (Table 4).

Table 4: Cost/variance trade-off, $\kappa = 0.1\%$. The 5% CVaR of the loss is minimised at $n^* = 168$.

n	std(error)	mean cost	5% CVaR
3	0.9604	0.1068	3.1145
6	0.7210	0.1183	2.2855
12	0.5322	0.1329	1.6956
21	0.4173	0.1479	1.3599
42	0.3082	0.1730	1.0940
84	0.2379	0.2078	0.9435
168	0.2139	0.2581	0.9246
336	0.2335	0.3276	1.0063

Leland (1985). Leland’s adjusted volatility prices the cost buffer into the hedge,

$$\sigma_L = \sigma \sqrt{1 + \text{sign} \cdot \sqrt{\frac{2}{\pi}} \frac{\kappa}{\sigma \sqrt{\delta t}}}.$$

Hedging the short call at σ_L (with $\kappa = 0.2\%$, $n = 84$) improves the cost-adjusted P&L on *both* axes versus the plain σ — higher mean (less cost drag) and lower dispersion.

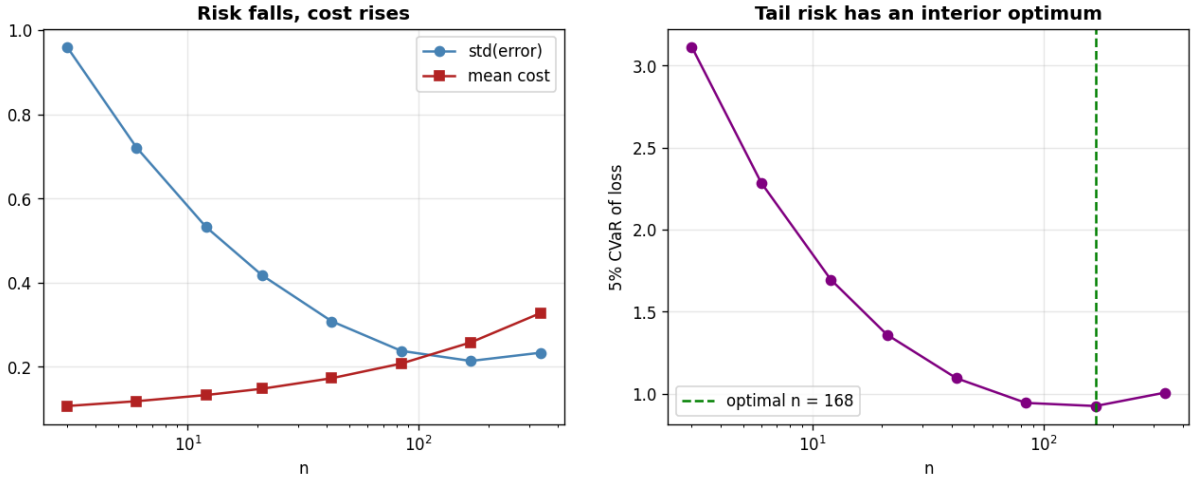


Figure 4: Left: dispersion falls while cost rises with n . Right: the 5% CVaR of the loss has an interior minimum at $n^* = 168$.

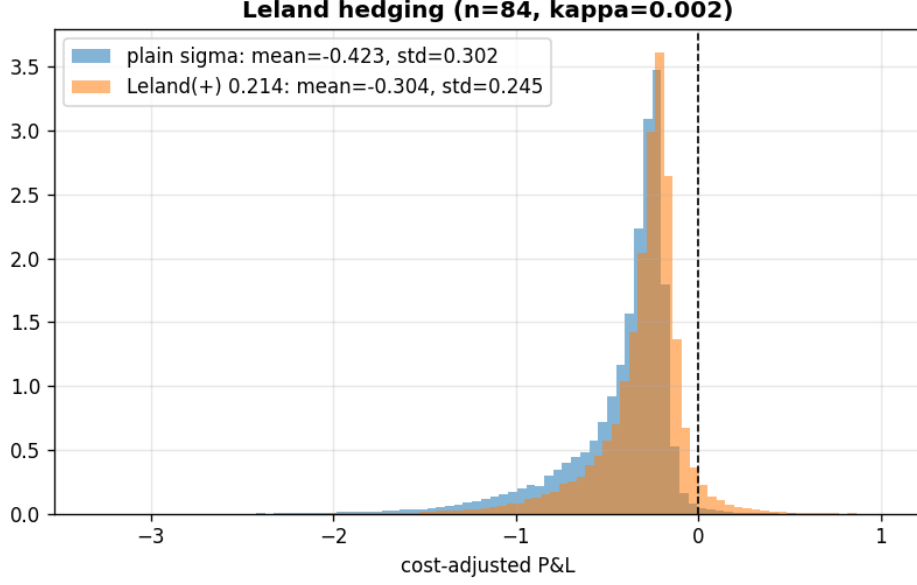


Figure 5: Leland’s adjusted-volatility hedge shifts the cost-adjusted P&L distribution right and tightens it.

7 Volatility misspecification and the gamma-P&L identity

This is the central result of option hedging. Hedging at an implied vol σ_h while the asset realises σ_r , the discrete hedging error of the short option is, to leading order,

$$\begin{aligned} \varepsilon &\approx -\frac{1}{2} \sum_k \Gamma_k S_k^2 \left[\left(\frac{\Delta S_k}{S_k} \right)^2 - \sigma_h^2 \delta t \right] \\ &= \underbrace{-\frac{1}{2} \sum_k \Gamma_k S_k^2 (\sigma_r^2 - \sigma_h^2) \delta t}_{\text{deterministic variance mismatch}} + \underbrace{(\text{mean-zero discretisation noise})}_{\sim (Z_k^2 - 1)}, \end{aligned}$$

where Γ_k is the Black–Scholes gamma at the *hedging* vol. The mean error is therefore driven by the variance mismatch $\sigma_r^2 - \sigma_h^2$, not the drift — the “robustness of Black–Scholes” result. Empirically the sign of the mean error tracks $\text{sign}(\sigma_r^2 - \sigma_h^2)$ exactly (Table 5).

Table 5: Mean call hedging error vs. hedging volatility, realised $\sigma_r = 0.20$, $n = 84$.

σ_h	0.14	0.17	0.20	0.23	0.26
mean error	−0.3481	−0.2102	+0.0004	+0.2735	+0.5972

The analytic decomposition matches the simulation engine to three decimals: with $\sigma_r = 0.20$, $\sigma_h = 0.25$, the engine mean error is +0.4844, the gamma-P&L sum is +0.4773 (correlation 0.994 path-by-path), and the deterministic mismatch term alone is +0.4777.

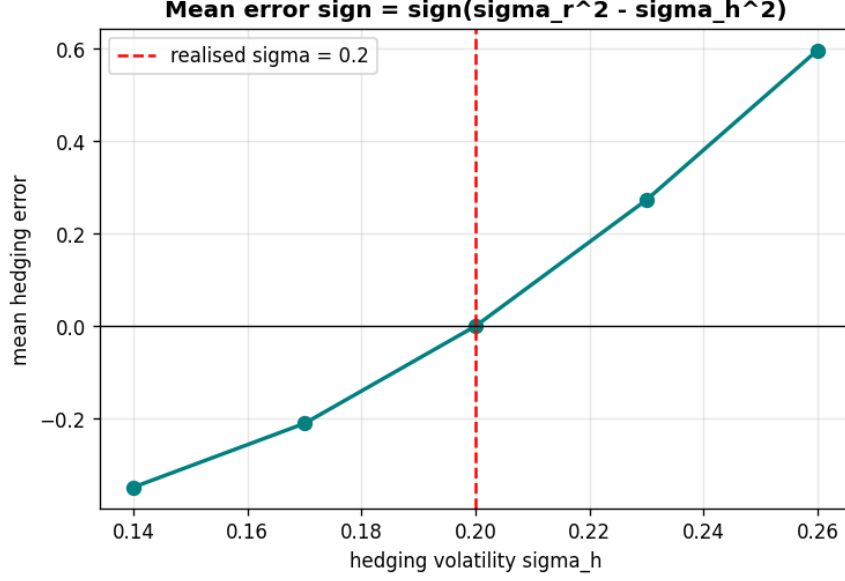


Figure 6: The mean hedging error changes sign exactly at $\sigma_h = \sigma_r$.

8 Greeks-based P&L attribution

Desks explain P&L with a second-order Taylor expansion, $dV \approx \theta dt + \Delta dS + \frac{1}{2}\Gamma dS^2$. Summed along the path ($n = 252$) it reproduces the option's value change to a tiny residual: the component means are θ -decay -2.79 , delta $+2.43$, gamma $+0.80$, giving a predicted $dV = +0.4339$ against an actual $+0.4335$, with residual standard deviation 0.018 against an actual-change std of 9.6 — the Greeks explain the P&L to $\sim 0.2\%$.

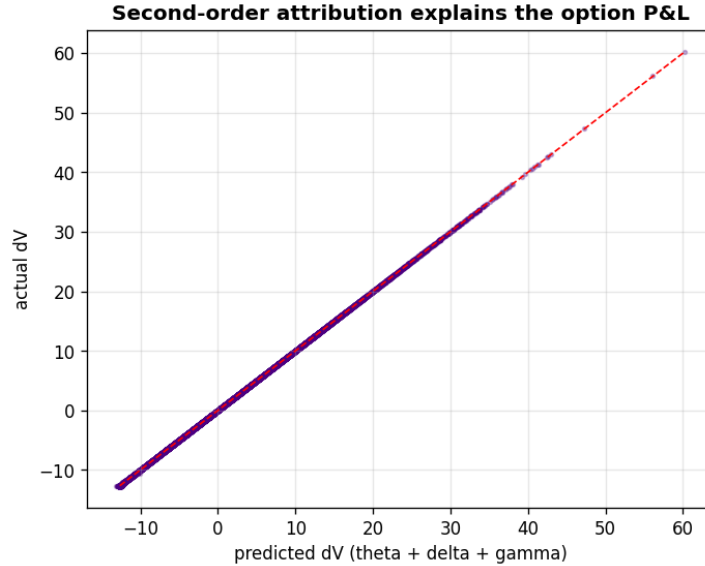


Figure 7: Predicted (theta + delta + gamma) versus actual value change: the points lie on the diagonal.

9 Model risk

Black–Scholes is a model, not reality. Keeping the same BS-delta hedge but simulating the underlying under *Heston* (stochastic volatility) and *Merton* (jump diffusion) degrades the hedge

sharply (Table 6). Jumps in particular break delta hedging and produce a fat-tailed loss distribution that no rebalancing frequency can control.

Table 6: BS-delta hedge ($n = 84$) under model misspecification: dispersion and tail risk of the hedging error.

underlying model	std	5% VaR	5% CVaR	min
GBM (correct)	0.2058	0.3252	0.5476	-2.16
Heston (stoch. vol)	0.6715	1.6490	2.4783	-7.40
Merton (jumps)	1.4020	1.4417	5.0280	-22.15

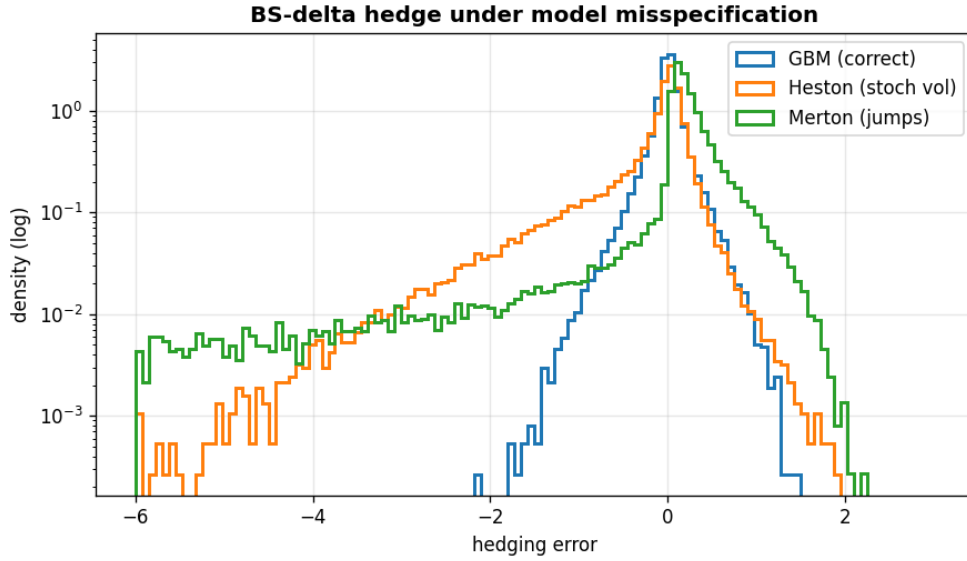


Figure 8: Hedging-error densities (log scale): the BS hedge is tight under GBM but develops heavy tails under stochastic volatility and especially jumps.

10 Conclusion

Discrete delta hedging of a vanilla call converges to perfect replication as the rebalancing frequency grows, with the error *dispersion* — not its mean, which is exactly zero — scaling as $n^{-1/2}$. That idealised picture is qualified in every realistic direction studied here: the dispersion depends on the drift; discontinuous (digital) payoffs converge far more slowly; transaction costs cap the useful frequency and make Leland’s adjustment worthwhile; the dominant risk is volatility misspecification, governed analytically by the gamma-P&L identity; and stochastic volatility or jumps leave fat-tailed residual risk no frequency can remove. Throughout, reporting Monte–Carlo standard errors is what separates a resolved effect from sampling noise.

Reproducibility. All numbers and figures are produced by the `deltahedge` package (exact GBM, $N = 50,000$, `numpy default_rng` seed 20,240,614) and pinned by a `pytest` regression suite. To reproduce:

```

pip install -e ".[dev]" && pytest &&
jupyter nbconvert -to notebook -execute -inplace
notebooks/Delta_Hedging_Analysis.ipynb

```